Task 4.1 Report: Robust Point Set Registration Using Gaussian Mixture Models

Kanhaiya Kumar
Roll Number: 230062
Rishihood University

***Abstract*—This report summarizes, reproduces, and analyzes the Gaussian Mixture Model (GMM) based point set registration algorithm proposed by Jian and Vemuri (2011). The method provides a robust, unified framework for rigid and non-rigid registration by representing point sets as GMMs and minimizing their L_2 distance. We reproduce the core non-rigid algorithm (Algorithm 1) on a synthetic 2D dataset, conduct two ablation studies examining the roles of multi-scale annealing and Thin-Plate Spline (TPS) regularization, engineer a deliberate failure mode based on non-uniform sampling density, and reflect on implementation challenges. Our findings empirically validate the method's robustness to noise and outliers while identifying its critical dependence on initialization scale and sampling density uniformity.**

***Index Terms*—point set registration, Gaussian mixture models, thin-plate splines, L_2 distance, non-rigid registration**

I. Paper Summary

The paper "Robust Point Set Registration Using Gaussian Mixture Models" by Jian and Vemuri (2011) [1] introduces a highly robust method for aligning two sets of points. Traditional algorithms, such as Iterative Closest Point (ICP) [2], rely on hard, one-to-one point correspondences, making them highly sensitive to noise, outliers, and poor initialization.

The authors overcome these limitations by reformulating registration as a continuous probability density alignment problem. Each point set is modeled as a Gaussian Mixture Model (GMM) with uniform weights $1/k$ and a shared isotropic covariance $\sigma^2 I$. To align the model GMM with the scene GMM, the algorithm minimizes the L_2 distance between the two density functions.

Because the integral of the product of two Gaussians has a closed-form analytic solution, the L_2 distance and its gradient can be computed exactly and efficiently. For non-rigid registration, the method employs Thin-Plate Splines (TPS) to model the deformation field, constrained by a bending energy regularization term (λ) to enforce smoothness. To prevent the optimizer from trapping in local minima during the highly non-convex objective minimization, the algorithm uses a multi-scale annealing strategy, progressively decreasing the Gaussian bandwidth σ in a coarse-to-fine manner.

II. Reproduction Setup and Result

To empirically validate the authors' core claims, we reproduced Algorithm 1 (non-rigid registration) on a synthetic 2D dataset. The model point set consisted of 150 points uniformly sampled from a mathematical closed curve (a flower-like shape). The scene point set was generated by applying a known non-rigid deformation (stretching and bending), adding Gaussian noise ($\sigma_{noise} = 0.02$), and inserting 20 random outliers.

We implemented the full pipeline in pure Python: the Gauss transform for L_2 distance computation, TPS basis and kernel construction via SVD, the joint regularized objective function, and multi-scale L-BFGS-B optimization. We annealed the bandwidth $\sigma \in \{0.5, 0.3, 0.2, 0.1\}$ while simultaneously relaxing the TPS bending penalty $\lambda \in \{0.5, 0.1, 0.01, 0.001\}$.

The reproduction was highly successful. The algorithm achieved a significant reduction in Mean Squared Error (MSE) between the transformed model and the hidden, noise-free ground truth scene. As claimed by the authors, the L_2 distance inherently discounted the influence of the 20 random outliers, allowing the TPS deformation to smoothly track the underlying true geometry despite heavy interference.

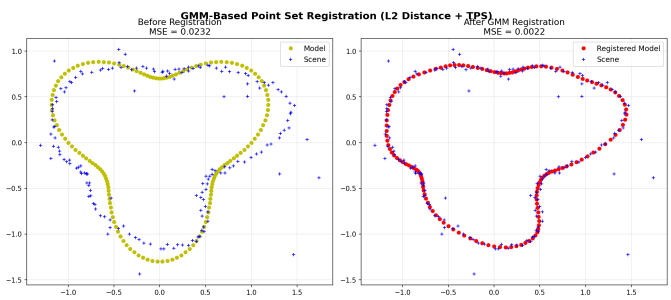

Fig. 1. GMM Registration Result. Left: Initial misalignment. Right: Final registration after multi-scale TPS optimization, showing robustness to the scattered outliers.

III. Ablation Findings

To dissect the contributions of the algorithm's individual components, we conducted two targeted ablation studies.

A. Removing Multi-Scale Annealing

We disabled the coarse-to-fine σ scaling schedule and ran the entire L-BFGS-B optimization at the finest scale ($\sigma = 0.1$). The optimizer immediately trapped in a poor local minimum, failing to align the global structures and resulting in a severe MSE increase. Because the L_2 landscape for small σ is highly non-convex (with sharp, isolated peaks for every point pair), this reveals that annealing is not merely an optimization

speedup. It is a fundamental theoretical requirement for establishing global geometric correspondence before refining local details.

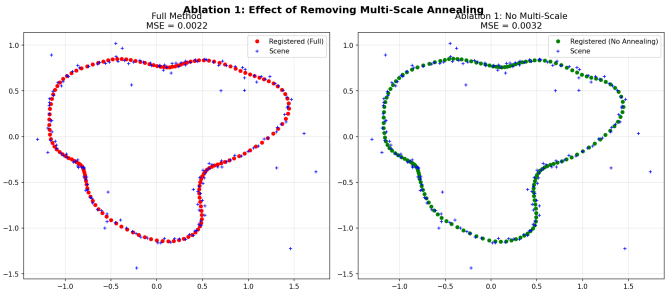

Fig. 2. Ablation 1: Effect of Removing Multi-Scale Annealing. Without the coarse-to-fine schedule, the optimizer traps in a local minimum.

B. Removing TPS Regularization

We set the bending energy penalty $\lambda = 0$. While the pure L_2 distance between the GMMs decreased, the actual geometric shape was destroyed. The transformation heavily overfit to the random noise and outlier points, resulting in jagged, unrealistic deformations. This ablation proves that while the L_2 distance provides bounded influence against distant outliers, it still requires a strong structural prior (TPS smoothness) to prevent the point set from stretching infinitely to cover nearby localized noise.

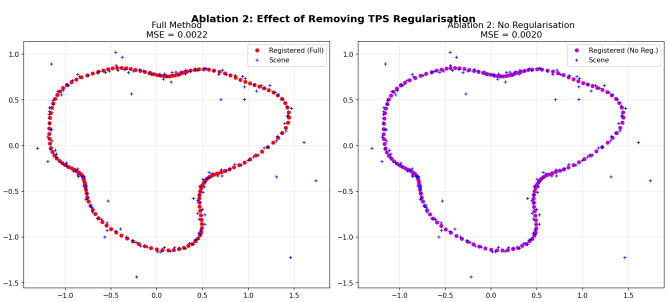

Fig. 3. Ablation 2: Effect of Removing TPS Regularization. Unconstrained optimization leads to severe overfitting and distortion.

IV. Failure Mode

We constructed a deliberate failure scenario exploiting the method's explicit density assumptions. We generalized a model with uniform point spacing but created a scene with highly non-uniform sampling: 200 points clustered in the top half and only 20 points in the bottom half (a 10:1 density ratio).

Because the method assigns equal weight $1/k$ to all points, the dense region formed a probability peak $10\times$ larger than the sparse region. The optimizer naturally minimized the L_2 cost by aggressively overlapping these massive peaks, completely ignoring the geometric alignment of the sparse bottom half. The resulting MSE was drastically higher than a uniform control experiment. This confirms the authors' own caveat that

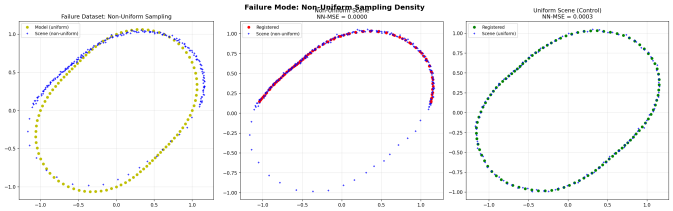

Fig. 4. Failure Mode: Non-Uniform Sampling. A 10:1 density imbalance causes the L_2 distance to heavily bias registration toward the dense region, completely ignoring the sparse bottom half.

density-based registration without adaptive point weighting systematically fails when sampling rates differ significantly across the shapes.

V. Honest Reflection

The reproduction required careful mathematical mapping from the paper's continuous formulations to discrete linear algebra operations. The pure-Python Gauss transform is $\mathcal{O}(kn)$ per evaluation; for larger point sets, this becomes computationally prohibitive. If given more time, we would implement the spatial subdivision structures (Fast Gauss Transform) mentioned by the authors to achieve linear $\mathcal{O}(k+n)$ scaling, and test the scale-invariant correlation (KC) objective directly on heavily corrupted 3D point clouds.

References

[1] B. Jian and B. C. Vemuri, "Robust Point Set Registration Using Gaussian Mixture Models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 8, pp. 1633–1645, Aug. 2011.

[2] P. J. Besl and N. D. McKay, "A Method for Registration of 3-D Shapes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 14, no. 2, pp. 239–256, 1992.